

CHAPTER 5

RANDOM VARIABLES: UNIVARIATE CASE

5.1 INTRODUCTION

In the first four chapters, we introduced the concept of sample space, identified events with subsets of this space, and developed some techniques of evaluating probabilities of events.

Random variables, each defined as numerical function on some sample space $\mathcal{S}$, will initially be regarded merely as useful tools for describing events. Then, if X stands for a random variable, inequality such as $X \leq t$ determines a set of all outcomes s in $\mathcal{S}$ satisfying the condition $X(s) \leq t$. We will postulate¹⁰ that $\{X \leq t\}$ is an event for each t . This way we gain a powerful tool for describing events in addition to the techniques used thus far, such as specifying subsets of sample space $\mathcal{S}$ by listing their elements or by providing a verbal description.

(*) However, in Chapter 1, it was repeatedly stressed that the concept of sample space is, to a large extent, a mathematical construction, and that one can have several sample spaces, all of them equally acceptable for the same phenomenon. The problem then is how to reconcile the inherent lack of uniqueness of the sample space used to describe a given phenomenon with the idea of a random variable being a function defined on the sample space. At first glance, it would appear that the concept of a random variable, being based on another concept (sample space $\mathcal{S}$) that allows subjective freedom of choice, must itself be tainted by subjectivity and therefore be of limited use.

¹⁰The reader may wonder why we use the term *postulate* here: the set $\{s \in \mathcal{S} : X(s) \leq t\}$ is a subset of $\mathcal{S}$ and consequently, is an event. However, as explained in Section 2.4, in the case of nondenumerable sample spaces $\mathcal{S}$ we may be unable to define probability P on the class of *all* subsets of $\mathcal{S}$. We must therefore restrict considerations to some class of subsets of $\mathcal{S}$, forming a σ -field, say $\mathcal{A}$, of events. In defining random variables, we postulate that $\{s \in \mathcal{S} : X(s) \leq t\} \in \mathcal{A}$ for every t . In accepted terminology, we say that random variables X are functions on $\mathcal{S}$ that are *measurable* with respect to the σ -field $\mathcal{A}$. From now on we will always tacitly assume that each set of the form $\{X \leq t\}$ is an event, and therefore it is permissible to speak of its probability.

Specifically, suppose that there are several sample spaces, say $\mathcal{S}, \mathcal{S}', \dots$, with the associated probability measures $P, P', \dots$ that are equally acceptable for describing a given phenomenon of interest, and let $X, X', \dots$ be random variables defined, respectively, on $\mathcal{S}, \mathcal{S}', \dots$.

As we will explain, the random variables that are useful for analyzing a given phenomenon must satisfy some “invariance” principle that make them largely independent of the choice of the sample space. The general idea is that a random variable is exactly what the name suggests: *a number depending on chance*. Phrases of this sort were commonly used as “definitions” of the concept of random variable before the probability theory was built on the notion of sample space. What we want to achieve is to define random variables in a way that makes them associated with the phenomenon studied rather than with a specific sample space, hence invariant under the choice of sample space. This is accomplished by introducing the following principle: (*)

Invariance Principle for Random Variables

Suppose that X' and X'' are two random variables defined on two sample spaces $\mathcal{S}'$ and $\mathcal{S}''$ used to describe the same phenomenon. We say that these two random variables are *equivalent* if for every t the event $\{X' \leq t\}$ occurs if and only if the event $\{X'' \leq t\}$ occurs, and moreover if these events have the same probability.

Clearly, the equivalence of random variables defined above satisfies the logical requirements for relation of equivalence: reflexivity, symmetry, and transitivity. We can therefore consider equivalence classes of random variables and speak of representatives of these classes. Random variables are particularly useful in describing and analyzing random phenomena: they do not depend on a particular choice of a sample space; hence they are free of the subjectivity and arbitrariness involved in the selection of $\mathcal{S}$.

5.2 DISTRIBUTIONS OF RANDOM VARIABLES

Although formally each random variable is a numerical function on some sample space $\mathcal{S}$, we usually suppress the dependence on elements of $\mathcal{S}$ in notation, referring to it only when needed to avoid confusion. As symbols for random variables, we will use capital letters from the end of the alphabet, X, Y, Z , possibly with subscripts.

With every random variable, we will associate its *probability distribution*, or simply *distribution*.

Definition 5.2.1 By a *distribution* of the random variable X we mean the assignment of probabilities to all events defined in terms of this random variable, that is, events of the form $\{X \in A\}$, where A is a set of real numbers. $\square$

Formally, the event above is a subset of the sample space $\mathcal{S}$. We have

$$\{X \in A\} = \{s \in \mathcal{S} : X(s) \in A\} \subset \mathcal{S}.$$

The basic type of events that we will consider is an interval, that is,

$$\{a < X < b\}, \quad \{a \leq X < b\}, \quad \{a \leq X \leq b\}, \quad \{a < X \leq b\}, \quad (5.1)$$

where $-\infty \leq a \leq b \leq \infty$.

If we can compute probability of each of the events (5.1) for all $a \leq b$, then using the rules from Chapter 2, we can compute probabilities of more complex events. For example,

$$\begin{aligned} P[\{a < X \leq b\}^c] &= P\{X \leq a \text{ or } X > b\} = P\{X \leq a\} + P\{X > b\} \\ &= P\{X \leq a\} + 1 - P\{X \leq b\}. \end{aligned}$$

Similarly, for the probability of the intersection $\{a < X \leq b\} \cap \{c < X \leq d\}$, the answer depends on the mutual relationship between a, b, c , and d (in addition to assumptions $a \leq b$ and $c \leq d$). If, for instance, $a < c < b < d$, then the event in question reduces to $\{c < X \leq b\}$, and so on.

Actually, it turns out that it is sufficient to know the probabilities of only one type of event (5.1) in order to determine the probabilities of remaining types. We will prove one such relation; the proofs of others are similar.

Theorem 5.2.1 *The probabilities of the events of the form $\{a < X \leq b\}$ for all $-\infty \leq a \leq b \leq \infty$ uniquely determine the probabilities of events of the form $\{a < X < b\}$, $\{a \leq X \leq b\}$, and $\{a \leq X < b\}$.*

Proof: We have

$$\begin{aligned} \{a < X < b\} &= \bigcup_n \left\{ a < X \leq b - \frac{1}{n} \right\}, \\ \{a \leq X \leq b\} &= \bigcap_n \left\{ a - \frac{1}{n} < X \leq b \right\}, \\ \{a \leq X < b\} &= \bigcap_n \bigcup_m \left\{ a - \frac{1}{n} < X \leq b - \frac{1}{m} \right\}. \end{aligned} \quad (5.2)$$

We will prove the last of these identities. Let s be a sample point belonging to the left-hand side so that $a \leq X(s) < b$. Then for every n , we have $a - 1/n < X(s)$. Also, if $X(s) < b$, then $X(s) \leq b - 1/m$ for some m (actually for all m sufficiently large). Thus, s belongs to the set $\bigcap_n \bigcup_m \{a - 1/n < X \leq b - 1/m\}$. On the other hand, if s belongs to the right-hand side of the last equality in (5.2), then we have

$$a - \frac{1}{n} < X(s) \leq b - \frac{1}{m}$$

for all n and for some m . Passing to the limit with $n \rightarrow \infty$, we obtain $a \leq X(s) \leq b - 1/m < b$, which means that s belongs to the set $\{a \leq X < b\}$. $\square$

Thus, each of the sets $\{a < X < b\}$, $\{a \leq X \leq b\}$, and $\{a \leq X < b\}$ can be represented as union or intersection (or both) of a sequence of events of the form $\{a_n < X \leq b_n\}$.¹¹

Let us observe that $\{a < X \leq b\} = \{X \leq b\} \setminus \{X \leq a\}$. Consequently,

$$P\{a < X \leq b\} = P\{X \leq b\} - P\{X \leq a\}$$

and the probabilities of all events of the form $\{a < X \leq b\}$ are determined by probabilities of events of the form $\{X \leq t\}$ for $-\infty < t < \infty$. This justifies the following important definition:

Definition 5.2.2 For any random variable X , the function of real variable t defined as

$$F_X(t) = P\{X \leq t\} \quad (5.3)$$

is called the *cumulative probability distribution function*,¹² or simply *cumulative distribution function (cdf)* of X . $\square$

¹¹In terminology introduced in Chapter 1, we could say that each of the three types of events above belongs to the smallest σ -field generated by the class of all events of the form $\{a < X \leq b\}$. Equivalently, we could say that the intervals on the real line of the kinds (a, b) , $[a, b)$, and $[a, b]$ belong to the smallest σ -field of sets of real numbers that contains all intervals of the form $(a, b]$.

¹²According to the footnote at the beginning of this chapter, the right-hand side of (5.3) is well defined for each t (i.e., $\{X \leq t\}$ is an event).

The following two examples will illustrate the concept of a cdf:

■ EXAMPLE 5.1

The experiment consists of shooting once at a circular target T of radius R . Assume that it is certain that the target will be hit, and that the probability of hitting a particular section A contained in T is given by $|A|/|T|$, where $|\cdot|$ is the area of the section. We will determine the cdf of a random variable X defined as the distance between the point of hitting and the center of the target.

SOLUTION. A natural sample space in this case is just T . Without loss of generality, we can put the center of coordinate system in the center of T so that sample points $s = (x, y)$ satisfy $x^2 + y^2 \leq R^2$. If the target is hit at point $s = (x, y)$, then $X = X(s) = \sqrt{x^2 + y^2}$ is the distance of the point of hit from the origin. Clearly, we have $0 \leq X \leq R$.

Now let $F_X(t) = P\{X \leq t\}$. Obviously, if $t > R$, then $F_X(t) = 1$, and if $t < 0$, then $F_X(t) = 0$. For t with $0 \leq t \leq R$, we have

$$\begin{aligned} F_X(t) &= P\{X \leq t\} \\ &= P\{\text{point } s \text{ falls in the circle of radius } t \text{ centered at the origin}\} \\ &= \frac{\pi t^2}{\pi R^2} = \left(\frac{t}{R}\right)^2. \end{aligned}$$

Thus

$$F_X(t) = \begin{cases} 0 & t < 0 \\ \left(\frac{t}{R}\right)^2 & 0 \leq t \leq R \\ 1 & t > R. \end{cases} \quad (5.4)$$

The graph of $F_X(t)$ is given in Figure 5.1.

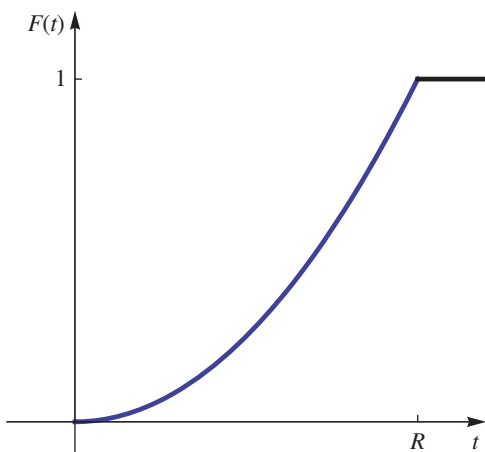


Figure 5.1 Cdf of the distance from the center of the target.

Remark A simple fact, useful in determining cdf's of random variables, is that if the random variable X is bounded from above, then $F_X(t) = 1$ for all $t > M$, where M is the upper

bound for X . Indeed, if $t > M$, then

$$\begin{aligned} F_X(t) &= P\{X \leq t\} \\ &= P\{X \leq M\} + P\{M < X \leq t\} = P\{X \leq M\} = 1. \end{aligned}$$

Similarly, if X is bounded from below, then $F_X(t) = 0$ for $t < m$, where m is the lower bound for X .

■ EXAMPLE 5.2

Let the experiment consist of three tosses of a coin, and let X be total number of heads obtained. The sample space consists of the eight sample points listed below, together with their associated values of X :

$\mathcal{S}$	X	$\mathcal{S}$	X
HHH	3	TTH	1
HHT	2	THT	1
HTH	2	HTT	1
THH	2	TTT	0

The random variable X satisfies the condition $0 \leq X \leq 3$ so that $F_X(t) = 0$ if $t < 0$ and $F_X(t) = 1$ if $t > 3$. Moreover, since X can take on only values 0, 1, 2, and 3, we have $P\{X \in A\} = 0$ for every set A that does not contain any of the possible values of X . Finally, simple counting gives $P\{X = 0\} = P\{X = 3\} = 1/8$ and $P\{X = 1\} = P\{X = 2\} = 3/8$. Thus for $0 \leq t < 1$ we write

$$\begin{aligned} F_X(t) &= P\{X \leq t\} = P\{X < 0\} + P\{X = 0\} + P\{0 < X \leq t\} \\ &= P\{X = 0\} = \frac{1}{8}. \end{aligned}$$

Similarly, if $1 \leq t < 2$, then

$$\begin{aligned} F_X(t) &= P\{X \leq t\} \\ &= P\{X < 0\} + P\{X = 0\} + P\{0 < X < 1\} \\ &\quad + P\{X = 1\} + P\{1 < X \leq t\} \\ &= P\{X = 0\} + P\{X = 1\} = \frac{1}{2}. \end{aligned}$$

Proceeding in a similar way with $2 \leq t < 3$, we get

$$F_X(t) = \begin{cases} 0 & \text{for } t < 0 \\ \frac{1}{8} & \text{for } 0 \leq t < 1 \\ \frac{1}{2} & \text{for } 1 \leq t < 2 \\ \frac{7}{8} & \text{for } 2 \leq t < 3 \\ 1 & \text{for } 3 \leq t. \end{cases}$$

The graph of $F_X(t)$ is given in Figure 5.2.

We will next investigate general properties of cdf's in some detail.

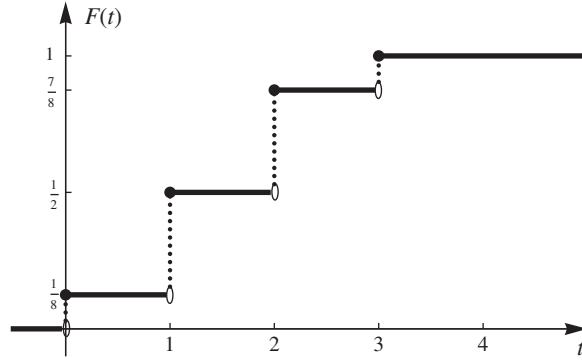


Figure 5.2 Cdf of the number of heads in three tosses of a coin.

Theorem 5.2.2 Every cdf F has the following properties:

- (a) F is nondecreasing.
- (b) $\lim_{t \rightarrow -\infty} F(t) = 0$, $\lim_{t \rightarrow \infty} F(t) = 1$.
- (c) F is continuous on the right.

Proof: Let F be the cdf of random variable X . To prove (a), let $t_1 < t_2$. Then

$$F(t_2) - F(t_1) = P\{X \leq t_2\} - P\{X \leq t_1\} = P\{t_1 < X \leq t_2\} \geq 0.$$

Consequently, $F(t_1) \leq F(t_2)$, as was to be shown.

To prove the second relation in (b), we have to show that $\lim_{n \rightarrow \infty} F(t_n) = 1$ for any sequence $\{t_n\}$ such that $t_n \rightarrow \infty$. Without a loss of generality, we can assume that $t_1 < t_2 < \dots \rightarrow \infty$. The events $\{X \leq t_1\}, \{t_1 < X \leq t_2\}, \{t_2 < X \leq t_3\}, \dots$ are disjoint, and their union is the whole sample space $\mathcal{S}$. By the second and third axioms of probability, we have

$$\begin{aligned} 1 &= P(\mathcal{S}) = P\{X \leq t_1\} + \sum_{k=2}^{\infty} P\{t_{k-1} < X \leq t_k\} \\ &= F(t_1) + \sum_{k=2}^{\infty} [F(t_k) - F(t_{k-1})] = F(t_1) + \lim_{n \rightarrow \infty} \sum_{k=2}^n [F(t_k) - F(t_{k-1})] \\ &= \lim_{n \rightarrow \infty} \{F(t_1) + [F(t_2) - F(t_1)] + \dots + [F(t_n) - F(t_{n-1})]\} \\ &= \lim_{n \rightarrow \infty} F(t_n), \end{aligned}$$

as was to be shown. The proof that $\lim_{t \rightarrow -\infty} F(t) = 0$ is analogous.

To prove (c), let $\{t_n\}$ be a sequence such that $t_1 > t_2 > \dots \rightarrow t^*$. We have to show that $\lim_{n \rightarrow \infty} F(t_n) = F(t^*)$.

The events $\{X \leq t_n\}$ form a monotonically decreasing sequence, with $\bigcap_{n=1}^{\infty} \{X \leq t_n\} = \{X \leq t^*\}$. So, by Theorem 2.6.1, we have

$$\begin{aligned} F(t^*) &= P\{X \leq t^*\} = P\left[\bigcap_{n=1}^{\infty} \{X \leq t_n\}\right] \\ &= P\left[\lim_{n \rightarrow \infty} \{X \leq t_n\}\right] = \lim_{n \rightarrow \infty} P\{X \leq t_n\} = \lim_{n \rightarrow \infty} F(t_n). \end{aligned}$$

□

One of the consequences of the foregoing properties of cdf's is that for every x

$$P\{X = x\} = F(x) - F(x-0), \quad (5.5)$$

where $F(x-0) = \lim_{h \downarrow 0} F(x-h)$ is the left-hand-side limit of F at x . Indeed, we have

$$\{X = x\} = \bigcap_{n=1}^{\infty} \{x - h_n < X \leq x\} \quad (5.6)$$

for every decreasing sequence of positive numbers h_n with $h_n \rightarrow 0$. Since the sets in the product (5.6) are decreasing, by the continuity property of probability,

$$\begin{aligned} P\{X = x\} &= \lim_{n \rightarrow \infty} P\{x - h_n < X \leq x\} = \lim_{n \rightarrow \infty} [F(x) - F(x - h_n)] \\ &= F(x) - \lim_{n \rightarrow \infty} F(x - h_n) = F(x) - F(x - 0). \end{aligned}$$

Sometimes it is necessary to identify values of a random variable corresponding to a given value of cdf. These values are called quantiles.

Definition 5.2.3 Let X be a random variable with cdf F , and let $0 < p < 1$. The *quantile of order p* (or *p th quantile*) ξ_p of X is defined as any solution of the simultaneous inequalities

$$P\{X \leq x\} \geq p, \quad P\{X \geq x\} \geq 1 - p. \quad (5.7)$$

The inequalities (5.7) are equivalent to

$$F(x) \geq p, \quad F(x - 0) \leq p. \quad \square$$

To illustrate this definition, we assume now that X is a random variable with continuous cdf $F(x)$. For any p with $0 < p < 1$ there exists¹³ a point ξ_p satisfying the relation $F(\xi_p) = p$. This means that $P\{X \leq \xi_p\} = p$. If F is continuous, then $P\{X \geq \xi_p\} = P\{X > \xi_p\} = 1 - F(\xi_p) = 1 - p$. So ξ_p satisfies (5.7). However, ξ_p is not always unique; F can satisfy the relation $F(x) = p$ for an interval of values x , and each of them can serve as ξ_p . For the case of random variables with continuous cdf, the condition (5.7) would then reduce to

$$P\{X \leq \xi_p\} = p, \quad P\{X \geq \xi_p\} = 1 - p, \quad (5.8)$$

with each of the relations (5.8) implying the other.

■ EXAMPLE 5.3

For the random variable X in Example 5.1, the quantiles can be determined from the relation $(\xi_p/R)^2 = p$; hence $\xi_p = R\sqrt{p}$. For instance, if $p = 0.25$, then $\xi_{0.25} = R/2$: chances of hitting the target at a distance less than $1/2$ of the radius are 0.25.

The reason for using inequalities in condition (5.7) instead of the simpler condition (5.8) lies in the fact that the equation $F(x) = p$ may have no solution, as illustrated by the following example:

■ EXAMPLE 5.4

Let X assume values 1, 2, and 3 with probabilities 0.2, 0.6, and 0.2, respectively. Then, $F(x) = 0.8$ for $2 \leq x < 3$, and $F(x) = 1$ for $x \geq 3$. Thus, the equation $F(x) = p$ is solvable only for $p = 0.2$ and $p = 0.8$. To find, for example, a $\xi_{0.5}$, we need to use

¹³The existence of such a point follows from the continuity (hence also the Darboux property) of function F ; it must assume every value between its lower bound 0 and upper bound 1.

(5.7), looking for points such that $P\{X \leq x\} \geq 0.5$ and $P\{X \geq x\} \geq 0.5$. The first inequality gives $x \geq 2$, while the second gives $1 - F(x - 0) \geq 0.5$. Hence $x \leq 2$, and both inequalities are satisfied only for $x = 2$. So we have $\xi_{0.5} = 2$.

Certain quantiles have special names. For example, $\xi_{0.5}$ is called the *median*, $\xi_{0.25}$ and $\xi_{0.75}$ are called *lower* and *upper* quartiles, respectively. Quantiles may also be called percentiles, p th quantile being the same as 100 p th percentile.

We have the following theorem:

Theorem 5.2.3 *If X is a random variable with continuous cdf and $0 < \alpha < \beta < 1$, then*

$$P\{\xi_\alpha \leq X \leq \xi_\beta\} = \beta - \alpha, \quad (5.9)$$

where ξ_α and ξ_β are any quantiles of order α and β , respectively.

Proof: The left-hand side of (5.9) equals $F(\xi_\beta) - F(\xi_\alpha - 0)$, which equals $F(\xi_\beta) - F(\xi_\alpha)$ by continuity of F , and hence equals $\beta - \alpha$ by the definition of quantiles. $\square$

Some comments about the concept of cdf are now in order. First, we defined the cdf of a random variable X using nonstrict inequality, that is, $F(t) = P\{X \leq t\}$. It is equally permissible to define cdf by the formula $F^*(t) = P\{X < t\}$. The only difference between F and F^* is that the latter is left continuous. Such a definition of the cdf actually appears throughout in Russian and Eastern European statistical literature.

The second comment concerns the sufficiency of conditions (a)–(c) in Theorem 5.2.2. To put it differently, if a function F satisfies conditions (a)–(c), does there exist a random variable X such that F is the cdf of X , that is, $F(t) = P\{X \leq t\}$?

The following theorem provides the answer. Interested readers could find the proof in advanced texts on probability, for example, Chow and Teicher (1997).

Theorem 5.2.4 *Any function F that satisfies conditions (a)–(c) of Theorem 5.2.2 is a cdf of some random variable.*

The importance of this theorem (as in the case of Theorem 2.6.3) is that it guarantees that some phrases, such as “let F be a cdf,” make sense, with no additional assumptions about F besides (a)–(c) of Theorem 5.2.2.

The next question, quite naturally, is: Do cdf’s determine uniquely the random variables associated with them? The answer here is negative: there are many random variables (associated with different phenomena, or even associated with the same phenomenon) that have the same cdf. Thus, it is a random variable that determines its cdf, and not conversely. To see why this is so, let us consider the following example:

■ EXAMPLE 5.5

Continuing Example 5.2, let us consider the experiment consisting of three tosses of a coin, and two random variables: X = total number of heads and Y = total number of tails. We have here, as before,

S	X	Y	S	X	Y
HHH	3	0	TTH	1	2
HHT	2	1	THT	1	2
HTH	2	1	HTT	1	2
THH	2	1	TTT	0	3

A simple count shows that $P\{X = k\} = P\{Y = k\}$ for all k , which implies that the cdf's $F_X(t)$ and $F_Y(t)$ are identical.

This example illustrates the fact that there exist two distinct variables X and Y defined on *the same* sample space $\mathcal{S}$, and that these two variables have the same distribution function. Obviously, if X and Y are defined as above, but refer to two distinct triplets of tosses, their distributions will still be the same.

On the cumulative distribution function of a random variable X , recall from Definition 5.2.1 that the distribution of random variable X is defined as an assignment of probabilities $P\{X \in A\}$, where A is a set of real numbers. The cumulative distribution function F_X then provides probabilities of intervals open on the left and closed on the right:

$$P\{X \in (a, b]\} = P\{a < X \leq b\} = F_X(b) - F_X(a). \quad (5.10)$$

The question now is: Does F_X also determine the probabilities $P\{X \in A\}$ for sets A other than intervals $(a, b]$, and if so, what is the class of these sets? Before answering, let us observe that the leftmost member of (5.10) can be written as $P\{s \in \mathcal{S} : X(s) \in (a, b]\}$, so that the symbol P refers to the subsets of sample space $\mathcal{S}$, as it should according to the definition of probability. On the other hand, the rightmost member depends only on a and b , and assigns a number to an interval $(a, b]$. We may therefore say that we are dealing with probability on sets of real numbers (at least on intervals). According to the footnote at the beginning of this chapter, the left-hand side is well defined: We know that events (sets on which P is defined) form a σ -field, and we also know that sets $\{X \leq t\}$ are events so that

$$\{a < X \leq b\} = \{X \leq b\} \setminus \{X \leq a\}$$

is also an event.

The following extension theorem is given without proof.

Theorem 5.2.5 *If F is a cumulative distribution function, then the function m_F defined on the class $\mathcal{B}_0$ of all intervals $(a, b]$, $-\infty \leq a \leq b \leq +\infty$, by the formula*

$$m_F(a, b] = F(b) - F(a) \quad (5.11)$$

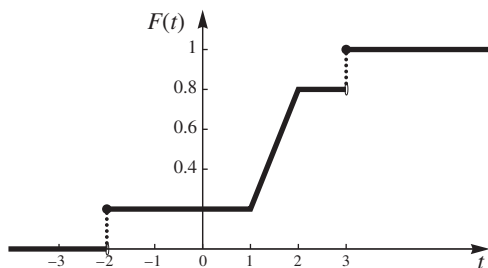
can be extended uniquely to a probability measure m_F on the smallest σ -field $\mathcal{B}$ containing $\mathcal{B}_0$.

PROBLEMS

5.2.1 In the statements below, F and G stand for cdf's of random variables X and Y , respectively. Classify each of the statements below as true or false: **(i)** If X is always strictly positive, then $F(t)$ is strictly positive for all t . **(ii)** If $F(37) = F(45)$, then $P(40 < X < 42) = P(43 < X < 44)$. **(iii)** If $Y = X + 3$, then $G(t) \leq F(t)$ for all t . **(iv)** If $G(17) - F(17) = 1$, then both X and Y are always less than 17. **(v)** If $G(17) - F(17) = 1$, then Y always exceeds X . **(vi)** If $G(17) \times F(17) = 1$, then $P(\max(X, Y) \leq 17) = 1$. **(vii)** If $G(17) \times F(17) = 0$, then $P(\min(X, Y) \leq 17) = 1$. **(viii)** If $|F(t) - G(t)| < \epsilon$ for all t , then $|X - Y| < \epsilon$. **(ix)** If $P(X \leq Y) = 1$, then $F(t) \geq G(t)$ for all t .

5.2.2 Figure 5.3 shows the cdf of a random variable X . Find: **(i)** $P(X = -2)$, $P(X = 0)$. **(ii)** $P(X \leq 3)$, $P(X < 3)$, $P(X < 0.13)$. **(iii)** $P(X > 2)$, $P(X > 2.79)$. **(iv)** $P(-1 < X \leq 0.7)$, $P(-2 \leq X < 1)$. **(v)** $P(1 \leq |X| \leq 2)$.

5.2.3 Let X be a random variable with cdf given by $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1 - 0.3e^{-\lambda x}$ for $x \geq 0$. Determine: **(i)** $P(X = 0)$. **(ii)** λ if $P(X \leq 3) = 3/4$. **(iii)** $P(|X| \leq 5)$ using results of (ii).

Figure 5.3 Cdf of random variable X .

5.2.4 Determine the medians and lower and upper quartiles for random variables with the following cdf's:

$$(i) F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ kx^3 & \text{for } 0 \leq x \leq 1/\sqrt[3]{k}, \quad k > 0 \\ 1 & \text{for } x > 1/\sqrt[3]{k}, \end{cases}$$

$$(ii) F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - \alpha e^{-x} & \text{for } x \geq 0. \end{cases} \quad (\text{consider all possible } \alpha)$$

5.2.5 A point is chosen at random from a square with side a . Let X be the distance from the selected point to the nearest corner of the square. Find and graph $F_X(x)$.

5.2.6 A coin of diameter d is dropped on a floor covered with square tiles with side length $D > d$. Let X be the number of tiles which intersect with the coin. **(i)** Find the distribution of X . **(ii)** Determine the median of X as a function of D and d .

5.2.7 Prove the first part of assertion (b) of Theorem 5.2.2.

5.3 DISCRETE AND CONTINUOUS RANDOM VARIABLES

Although the cdf of a random variable X provides all the information necessary to determine probabilities $P\{X \in A\}$ for a large class of sets A , there exist wide and practically important classes of random variables whose distributions may be described in simpler ways. Two such classes will be discussed in this section. Accordingly, we introduce the following definition:

Definition 5.3.1 A random variable X will be called *discrete*, if there exists a finite or countably infinite set of real numbers $U = \{x_1, x_2, \dots\}$ such that

$$P\{X \in U\} = \sum_n P\{X = x_n\} = 1. \quad (5.12)$$

□

■ EXAMPLE 5.6

Let $U = \{1, 2, \dots, n\}$ for some n , and let $P\{X = j\} = 1/n$ for $j \in U$. Condition (5.12) is satisfied. This example describes the selection (assumed fair) of the number of the winning lottery ticket, where n is the total number of tickets.

A discrete random variable with a finite set U of values, all of them having the same probability of occurrence, is called (discrete) *uniform*.

■ EXAMPLE 5.7 Binomial Distribution

The binomial distribution plays a central role in probability theory and statistical modeling, and it will often be used throughout this book. Here, we just introduce a definition and basic formulas.

Definition 5.3.2 The *binomial* random variable is defined as a total number of successes in n independent experiments, each experiment leading to success with probability p . $\square$

We have encountered special cases of binomial random variable in the preceding chapters (e.g., in analyzing the distribution of the “total number of heads in three tosses of a coin”). The set of possible values of X is $\{0, 1, \dots, n\}$, since the number of successes is an integer, and at best equals n (the number of trials) and at worst equals 0 (if all repetitions lead to failure). The probability of k successes and $n - k$ failures in any specific order $SFFS \dots S$ equals $p(1-p)(1-p)p \dots p = p^k(1-p)^{n-k}$. Since the probability of this string does not depend on its order, we obtain $P\{S_n = k\}$ by taking $p^k(1-p)^{n-k}$ as many times as there are different orders of k letters S and $n - k$ letters F . This number is $\binom{n}{k}$, since each such order is completely specified by selecting the set of locations for letter S among n slots. Thus,

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n, \quad (5.13)$$

and using the Newton binomial formula (3.16), we have

$$\sum_{k=0}^n P\{X = k\} = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = [p + (1-p)]^n = 1.$$

In the sequel, we shall use the symbol $\text{BIN}(n, p)$ to denote binomial distribution with parameters n and p . This way we say that X has distribution $\text{BIN}(n, p)$ or simply $X \sim \text{BIN}(n, p)$. We also let the individual probabilities in distribution $\text{BIN}(n, p)$ be denoted by $b(k; n, p)$ so that

$$b(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n. \quad (5.14)$$

In the examples above, the set U was finite, with $U = \{1, \dots, n\}$ in Example 5.6 and $U = \{0, 1, \dots, n\}$ in Example 5.7. In the example below, the set U is infinite.

■ EXAMPLE 5.8

Consider a sequence of independent tosses of a die. Let X be the number of tosses until the first ace. Clearly, X can assume values $1, 2, 3, \dots$, and the event $\{X = k\}$ occurs if the k th toss gives an ace (chances are $1/6$) and the first $k - 1$ tosses are all different from an ace [chances are $(5/6)^{k-1}$]. Thus

$$P\{X = k\} = \frac{1}{6} \left(\frac{5}{6}\right)^{k-1}, \quad k = 1, 2, \dots \quad (5.15)$$

We can easily check that

$$\sum_{k=1}^{\infty} P\{X = k\} = \frac{1}{6} \sum_{k=1}^{\infty} \left(\frac{5}{6}\right)^{k-1} = \frac{1}{6} \times \frac{1}{1 - \frac{5}{6}} = 1,$$

so condition (5.12) is satisfied.

This random variable is generally described as “waiting time for the first success” (in this case success being an ace). Distribution (5.15) is an example of *geometric* distribution, denoted GEO (p), which will be discussed in detail in Chapter 8.

■ EXAMPLE 5.9 Poisson Distribution

We say that the random variable X has *Poisson* distribution with parameter $\lambda > 0$, denoted POI (λ), if the possible values of X are nonnegative integers $0, 1, 2, \dots$ and

$$P\{X = k\} = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots \quad (5.16)$$

The Poisson distribution is often applied in practice, so we will study its properties in some detail in following sections. At present, observe that (5.16) is correctly defined, since

$$\sum_{k=0}^{\infty} P\{X = k\} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \times e^{\lambda} = 1.$$

In the examples above, the set U of possible values of X consisted of integers. This is the most frequent case, since typically discrete random variables represent counts involved in observing random phenomena. In other words, discrete random variables are typically obtained as “the number of ...” One should remember, however, that the definition allows any values in the set U , and not necessarily integers.

The distribution of a discrete random variable X is determined by the set $U = \{x_1, x_2, \dots\}$ of its possible values, and the assignment of probabilities $P\{X = x_i\}$ to all $x_i \in U$, the only condition being (5.12) (i.e., the sum of all probabilities must be 1). The function p_X , defined by the formula

$$p_X(x) = \begin{cases} P\{X = x\} & \text{if } x \in U \\ 0 & \text{if } x \notin U, \end{cases}$$

is often called the *probability mass function*, or the *probability function* of random variable X , or (in engineering-oriented texts) a probability *density* function. We will often use the first two of these terms. The term *density* can be confusing in this context, so we will reserve it for continuous random variables, discussed later in this section.

The cdf of a discrete random variable can now be easily determined. We have

$$F(t) = P\{X \leq t\} = \sum_{x_i \leq t} P\{X = x_i\}. \quad (5.17)$$

■ EXAMPLE 5.10

In the simplest cases, it is convenient to represent the distribution of a discrete random variable X as a double array of elements of U with corresponding probabilities, such as

Values	−3	$\frac{1}{2}$	5
Probability	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

We have here $U = \{-3, 1/2, 5\}$ and $P\{X = -3\} = 1/2$, $P\{X = 1/2\} = 1/3$, $P\{X = 5\} = 1/6$.

According to (5.17), $F(t) = 0$ for all $t < -3$, and for $-3 \leq t < 1/2$,

$$F(t) = P\{X \leq t\} = P\{X = -3\} = \frac{1}{2}.$$

For $1/2 \leq t < 5$, we have

$$P\{X \leq t\} = P\{X = -3\} + P\left\{X = \frac{1}{2}\right\} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

Finally, for $t \geq 5$, we have $P\{X \leq t\} = P\{X \leq 5\} = 1$.

It is worthwhile to point out that the cdf is defined for *all* real arguments, and not only for the possible values of the random variable. For example, one can find $F_X(2.27)$ if X is the result of a single toss of a regular die.

■ EXAMPLE 5.11

The random variable X in Example 5.8 was defined as the number of tosses of a die, up to and including the first ace. Here the cdf is as follows: First, since $X \geq 1$, we have $F(t) = 0$ for all $t < 1$. Second, if $k \leq t < k + 1$, where $k = 1, 2, \dots$, then we have

$$F(t) = P\{X \leq t\} = \sum_{j=1}^k P\{X = j\} = \sum_{j=1}^k \frac{1}{6} \left(\frac{5}{6}\right)^{j-1} = 1 - \left(\frac{5}{6}\right)^k.$$

In both examples, the cumulative distribution function is a step function, with cdf increasing at every point of the set U and constant between the steps. Although this is the property characterizing most discrete random variables, there also exist discrete random variables such that their cdf's are not constant on *any* interval, as shown in the example below.

■ EXAMPLE 5.12

Let U be the set of all rational numbers. It is well known that U is countable, that is, all elements of U can be arranged into a sequence $x_1, x_2, \dots$ (but U *cannot* be arranged in a sequence $x_1, x_2, \dots$ with $x_n < x_{n+1}$ for all n). Let X be a random variable such that $P\{X = x_n\} = 1/2^n, n = 1, 2, \dots$. Then, the cdf of X is not constant on any interval. Indeed, for $t_1 < t_2$ we have

$$F(t_2) - F(t_1) = \sum_{t_1 < x_n \leq t_2} P\{X = x_n\}, \quad (5.18)$$

and the right-hand side is positive, since there exists a rational number x_n between any two distinct real numbers t_1 and t_2 . Consequently, F increases between any two points.

Formula (5.18) shows how to calculate the probability of any interval $(t_1, t_2]$. Generally, for any set A , we have

$$P\{X \in A\} = \sum_{x_n \in A} P\{X = x_n\}.$$

■ EXAMPLE 5.13

The random variable X has the following distribution:

Value	-3	-2	0	1	2
Probability	$\frac{2}{9}$	$\frac{1}{9}$	p	$\frac{2}{9}$	p^2

and we want to find $P(|X + 1| > 1)$.

SOLUTION. First, we use (5.12) to find the value of p . We have here

$$\frac{2}{9} + \frac{1}{9} + p + p^2 + \frac{2}{9} = 1,$$

which gives the equation $p^2 + p - 4/9 = 0$. The solutions are $p = 1/3$ and $p = -4/3$, of which only the first is admissible as a probability. Consequently $P\{X = 0\} = 1/3$, $P\{X = 1\} = 1/9$ and

$$\begin{aligned} P\{|X + 1| > 1\} &= P\{X = -3\} + P\{X = 1\} + P\{X = 2\} \\ &= \frac{2}{9} + \frac{2}{9} + \frac{1}{9} = \frac{5}{9}. \end{aligned}$$

Let us now introduce another large class of random variables.

Definition 5.3.3 The random variable X will be called *continuous* if there exists a nonnegative function f , called the *density* of X , such that

$$F_X(t) = \int_{-\infty}^t f(x) dx \quad (5.19)$$

for $-\infty < t < \infty$. □

■ EXAMPLE 5.14 Uniform Distribution

Let $A < B$, and let the density $f(x)$ have the form:

$$f(x) = \begin{cases} 0 & \text{if } x < A \\ c & \text{if } A \leq x \leq B \\ 0 & \text{if } x > B. \end{cases} \quad (5.20)$$

If $t < A$, then the integrand in (5.19) is 0, so $F(t) = 0$. For $A \leq t \leq B$, we have

$$F(t) = \int_{-\infty}^A f(x) dx + \int_A^t f(x) dx = \int_{-\infty}^A 0 dx + \int_A^t c dx = c(t - A).$$

Finally, if $t > B$, then

$$F(t) = \int_{-\infty}^A f(x) dx + \int_A^B f(x) dx + \int_B^t f(x) dx = \int_A^B c dx = c(B - A).$$

This means that $F(t)$ is of the form presented in Figure 5.4. Clearly, since $\lim_{t \rightarrow \infty} F(t) = 1$ by Theorem 5.2.2, we must have $c(B - A) = 1$; hence $c = 1/(B - A)$. In other words, if function (5.20) is a density of a random variable, then c is uniquely determined by the length of the interval $[A, B]$. Thus, the density of the distribution (continuous) *uniform* on $[A, B]$, denoted $U[A, B]$, is

$$f(x) = \begin{cases} 0 & \text{if } x < A \\ \frac{1}{B - A} & \text{if } A \leq x \leq B \\ 0 & \text{if } x > B. \end{cases} \quad (5.21)$$

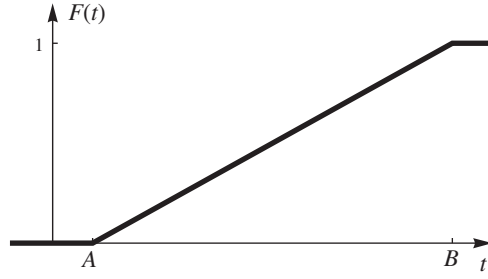


Figure 5.4 Cdf of distribution uniform on $[A, B]$.

In general, by letting $t \rightarrow \infty$ in (5.19), we see that every density function f must satisfy the relation

$$\int_{-\infty}^{+\infty} f(x) dx = 1. \quad (5.22)$$

Also, it follows from (5.19) that $F(t)$ is a continuous function, and consequently for every x ,

$$P(X = x) = F(x) - F(x - 0) = 0. \quad (5.23)$$

Formula (5.19) combined with (5.23) leads to the following theorem:

Theorem 5.3.1 *If random variable X has density f , then for all $a < b$ we have*

$$\begin{aligned} P\{a \leq X \leq b\} &= P\{a < X \leq b\} = P\{a < X < b\} \\ &= P\{a \leq X < b\} = F(b) - F(a) = \int_a^b f(x) dx. \end{aligned}$$

This formula implies that in dealing with continuous random variables, one can afford the luxury of being sloppy in handling inequalities. In particular, one can treat an event such as $\{X \leq a\}$ as (equivalent to) the complement of the event $\{X \geq a\}$, and so on. This is in sharp contrast with discrete random variables, where the events $\{X \leq a\}$ and $\{X < a\}$ may have different probabilities.

A question arises as to the extent to which the cdf of a continuous random variable determines its density. Formula (5.19) suggests that we have

$$F'(t) = f(t), \quad (5.24)$$

and consequently, since F is a nondecreasing function (by Theorem 5.2.2), we must have

$$f(t) \geq 0. \quad (5.25)$$

In fact formulas (5.24) and (5.25) need not be valid for all points t . The reason is that the density f determines the probabilities of intervals through integration. This means that f is not defined uniquely. Indeed, the two densities f_1 and f_2 —which differ only at a single point or on a finite set of points—will satisfy the condition

$$\int_a^b f_1(x) dx = \int_a^b f_2(x) dx$$

for all a and b . The same will be true if f_1 and f_2 differ on some set of measure zero.

Consequently, we may only claim that formulas (5.24) and (5.25) are valid almost everywhere, that is, except on a set of measure zero.

■ EXAMPLE 5.15

The random variable X , with density given in Example 5.14, is called *uniform* on interval $[A, B]$. It is clear, however, that if we modify the definition of f at boundaries A and B (e.g., by putting $f(A) = f(B) = 0$), the cdf will remain unchanged. A particular consequence of this observation is as follows: in real situations, we often deal with discontinuous densities defined by different formulas in different intervals. In such cases, it does not matter how the density is defined at the endpoints.

The last remark is true in regard to calculating probabilities. One should nevertheless be careful with the interpretation of density. For example, since for $\Delta x > 0$, we have

$$P\{x \leq X \leq x + \Delta x\} = \int_x^{x+\Delta x} f(t) dt,$$

we can approximate the last integral, for *continuous densities* f , by $f(x)\Delta x$. This leads to an “engineer’s” interpretation, according to which $f(x) dx$ is the “probability that random variable X will assume the value in the infinitesimal interval $(x, x + dx)$.” Here one has to be careful not to apply such an interpretation at points where f is not continuous. For instance, in the Example 5.14, we had $f(B) = 1/(B - A)$; hence the probability of X assuming a value in the interval $[B, B + h]$ for some small $h > 0$ may be taken as $h/(B - A)$. But this value is positive, whereas in fact the probability of X exceeding B by *any* amount is zero.

Let us now introduce an important type of continuous distributions called an exponential distribution.

Definition 5.3.4 (Exponential Distribution) The distribution with the density

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0, \end{cases} \quad (5.26)$$

where $\lambda > 0$, will be called *exponential* with parameter λ , denoted $\text{EXP}(\lambda)$. □

To check that (5.26) is indeed a density, we write

$$\int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{+\infty} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_0^{\infty} = 1.$$

According to the remark made in Example 5.15, the value of the density at $x = 0$ plays no role. Thus, we could have defined $f(x)$ in Definition 5.3.4 as $\lambda e^{-\lambda x}$ for $x \geq 0$ and 0 for $x < 0$.

■ EXAMPLE 5.16

Let us compute $P\{|X - 1/2| > 1/4\}$ for a random variable X that has $\text{EXP}(2)$ distribution.

SOLUTION. Since the density determines probabilities through integration,

$$\begin{aligned}
 P\{|X - 0.5| > 0.25\} &= \int_{|x-0.5|>0.25} f(x) \, dx \\
 &= \int_{x<0.25} f(x) \, dx + \int_{x>0.75} f(x) \, dx \\
 &= \int_{-\infty}^0 0 \, dx + \int_0^{0.25} 2e^{-2x} \, dx + \int_{0.75}^{\infty} 2e^{-2x} \, dx \\
 &= 1 - e^{-0.5} + e^{-1.5} = 0.6166.
 \end{aligned}$$

Example 5.16 should serve as a warning. It happens often that density is defined by a formula consisting of several parts, like (5.26). In calculating probabilities, one has to integrate f over some set, and special care should be taken to use the proper part of the formula for f on appropriate parts of the domain of integration.

We will always choose a version of density that is “most regular.” In particular, this means choosing a continuous or piecewise continuous version if possible. When density is discontinuous, the choice of values at breakpoints is irrelevant.

■ EXAMPLE 5.17 Normal Distribution

One of the most common distributions encountered in statistics, useful in modeling real-life phenomena, is the *normal* distribution, with density defined by

$$f(x) = f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, \quad (5.27)$$

where μ and $\sigma > 0$ are two parameters whose interpretation will be given later. The distribution (5.27) is usually denoted $N(\mu, \sigma^2)$. The distribution $N(0, 1)$, with density

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (5.28)$$

is called *standard normal*.

We will show that the function (5.27) is a density; that is, it integrates to 1. Indeed, letting $z = (x - \mu)/\sigma$, so that $\sigma \, dz = dx$, we have

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-(x-\mu)^2/2\sigma^2} \, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-z^2/2} \, dz.$$

Next, we need to prove that a standard normal density (5.28) integrates to 1.

The function $e^{-x^2/2}$ does not have an antiderivative that can be written in a closed form. Thus, to compute the integral

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2/2} \, dx,$$

we apply a trick and compute I^2 :

$$\begin{aligned}
 I^2 &= \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2/2} \, dx \right) \times \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-y^2/2} \, dy \right) \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} \, dx \, dy.
 \end{aligned} \quad (5.29)$$

Using polar coordinates (r, θ) , with $x = r \cos \theta$, $y = r \sin \theta$, we write the Jacobian of the transformation as

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

Then, we change the variables in (5.29) to obtain

$$I^2 = \frac{1}{2\pi} \int_0^{2\pi} d\theta \int_0^\infty r e^{-r^2/2} dr = 1;$$

hence $I = 1$, as was to be shown.

As already mentioned, the density function $\varphi(x)$ of standard normal distribution (as well as any other normal density function $f(x; \mu, \sigma)$) does not have an antiderivative expressible in closed form. Therefore, to find probabilities for normal random variables, one needs to use available statistical software.

Suppose that we have to find

$$P\{a \leq X \leq b\} = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx, \quad (5.30)$$

for random variable $X \sim N(\mu, \sigma^2)$. In the case of any normal distribution, probabilities can be obtained using probabilities of the standard normal distribution. To see why, consider the probability (5.30) and use the change of variable $z = (x - \mu)/\sigma$ in the integral. We have $dx = \sigma dz$; hence, after substitution to (5.30) we obtain

$$P\{a \leq X \leq b\} = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2} dz = \Phi(z_2) - \Phi(z_1),$$

where $z_1 = (a - \mu)/\sigma$, $z_2 = (b - \mu)/\sigma$, and Φ is the cdf of the standard normal distribution. One may also use the fact that $\varphi(z)$ is symmetric about 0 so that

$$\Phi(-z) = 1 - \Phi(z). \quad (5.31)$$

The procedure is illustrated by the following example:

■ EXAMPLE 5.18

Suppose that $X \sim N(-0.7, 4)$ and that U is defined as an integer nearest to X . Find $P(U = -1)$.

SOLUTION. In this case, $\mu = -0.7$ and $\sigma^2 = 4$; hence $\sigma = 2$. Consequently,

$$\begin{aligned} P(U = -1) &= P(-1.5 < X < 0.5) \\ &= \Phi\left(\frac{-0.5 - (-0.7)}{2}\right) - \Phi\left(\frac{-1.5 - (-0.7)}{2}\right) \\ &= \Phi(0.1) - \Phi(-0.4). \end{aligned}$$

Using (5.31) to reduce to positive arguments of Φ , we have

$$\begin{aligned} P(U = -1) &= \Phi(0.1) - (1 - \Phi(0.4)) = \Phi(0.1) + \Phi(0.4) - 1 \\ &= 0.5398 + 0.6554 - 1 = 0.1952, \end{aligned}$$

where the numerical values are obtained from the statistical software.

At the end of this section, it is necessary to point out that discrete and continuous random variables do not exhaust all possibilities. First, we may have practical situations of random variables of *mixed* type, partially discrete and partially continuous. Second, we may also have random variables that are neither continuous nor discrete. This second possibility may appear at first as a mathematical pathology of some sort; nevertheless, there are random variables occurring in practice that have such pathological distributions.

■ EXAMPLE 5.19

An example of a random variable of mixed type may be as follows. We purchase a piece of equipment (e.g., a light bulb). Its lifetime T is typically a continuous random variable in the sense that T can assume any value from some interval; thus, any particular value $T = t$ from this interval has probability zero. In addition the value $T = 0$ can be assumed with positive probability. In other words, the bulb may either be broken at the time of purchase (in which case $T = 0$), or it may break at some future time $t > 0$, at which time the event $\{T = t\}$ has probability zero. Consequently, the cdf of T is a function that equals 0 for all negative t , and is continuously increasing to 1 for positive t . At $t = 0$ the cdf is discontinuous, with $F(0) > 0$ being the probability of purchasing a broken light bulb.

Such mixtures of continuous and discrete distributions still do not exhaust all possibilities. This is illustrated by the following example, which is of theoretical interest:

■ EXAMPLE 5.20

We will now construct a cdf $F(t)$ that is continuous, increases from 0 to 1, and is such that $F'(t) = 0$, except on a set of measure 0. The latter condition excludes the existence of density; that is, if the density exists, then it equals $F'(t)$ almost everywhere. Thus, we have $f(t) = 0$ almost everywhere; hence, $\int_{-\infty}^{+\infty} f(t) dt = 0$, which means that F' is not a density.

The construction is based on the *Cantor set*. We let $F(t) \equiv 0$ for $t \leq 0$ and $F(t) \equiv 1$ for $t \geq 1$. Next, we let $F(t) \equiv 1/2$ for $1/3 \leq t < 2/3$. On middle parts of intervals $[0, 1/3]$ and $[2/3, 1]$, that is, for $1/9 \leq t < 2/9$ and for $7/9 \leq t < 8/9$, we let $F(t) \equiv 1/4$ and $F(t) \equiv 3/4$, respectively. This process is continued, and at each step, $F(t)$ is the average of values on neighboring intervals in the middle one-third of the “gap.” The total length of intervals in $[0, 1]$ where F is constant (hence where $F' = 0$) is

$$L = \frac{1}{3} + 2 \times \frac{1}{9} + 4 \times \frac{1}{27} + \cdots = \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = \frac{1}{3} \times \frac{1}{1 - \frac{2}{3}} = 1.$$

Moreover, one can easily show that F is continuous at each point. Thus, we have constructed a cdf of a random variable which is neither discrete nor continuous (it is called *singular*).

PROBLEMS

- 5.3.1** A die is biased in such a way that the probability of obtaining k dots ($k = 1, \dots, 6$) is proportional to k^2 . Which number of dots is more likely: odd or even?
- 5.3.2** You have five coins in your pocket: two pennies, two nickels, and a dime. Three coins are drawn at random. Let X be the total amount drawn (in cents). Find: (i) The distribution of X . (ii) $P(X \leq 10 | X \leq 15)$. (iii) The probabilities that two pennies are drawn, if it is known that $X \leq 11$.

- 5.3.3** Let X have the density $f(x) = Ce^{-0.4|x|}$, $-\infty < x < +\infty$ (such distribution is called *Laplace* or *double exponential*). Find C and then obtain: (i) $P(X > -2)$. (ii) $P(|X + 0.5| < 1)$.
- 5.3.4** Let X have EXP(1) distribution. Moreover, let $Y = [X]$ be the integer part of X , and let Z be the integer nearest to X . Find: (i) The distributions of Y and Z . (ii) $P(Y = Z)$. (iii) $P(Y = 3|Z = 4)$. (iv) $P(Z = 4|Y = 3)$. (v) $P(Y = 4|Z = 3)$. (vi) $P(Z = 3|Y = 4)$.
- 5.3.5** Let X have EXP(λ) distribution, and let Y and Z be defined as in Problem 5.3.4. (i) Find $P(Y = Z)$. (ii) Show that $P(Y = k|Z = k + 1) = P(Y = Z)$ for all $k = 0, 1, \dots$ (iii) Find $P(Z = k + 1|Y = k)$ for $k = 0, 1, \dots$.
- 5.3.6** Let X have EXP(λ) distribution. Show that for $s, t > 0$ the following *memoryless property* holds: $P\{X > s + t|X > s\} = P\{X > t\}$.
- 5.3.7** Let X_n be the difference (possibly negative) between the number of heads and the number of tails in n tosses of a coin. Find: (i) The distribution of X_4 . (ii) The cdf of X_4 at point $x = -0.6$. (iii) The probability that X_n is positive given that it is nonnegative for (a) $n = 4$ and (b) $n = 5$.
- 5.3.8** Let X have the density $f(x) = Cx$ for $0 \leq x \leq 1$, $f(x) = C(2 - x)/2$ for $1 < x \leq 2$, and $f(x) = 0$ otherwise. Find C and $F(x)$. Compute the following probabilities and show them on the graphs of $f(x)$ and $F(x)$: (i) $P(X \geq 3/2)$. (ii) $P(|X - 1| \leq 1/2)$. (*Hint: The problem can be solved without integration, just using simple geometry.*)
- 5.3.9** Let random variable X with the cdf F be uniformly distributed over the union of intervals $(0, a)$ and $(a + 2, b)$. Assuming that $F(4) = 0.2$ and $F(a + 1) = 0.25$, find: (i) a and b . (ii) $F(8.39)$. (iii) $P(3.01 \leq X \leq 9.14)$. (iv) The probability that among 10 values independently selected from this distribution, exactly two will be less than 4 and exactly three will be larger than 17.
- 5.3.10** Light bulbs of a certain type, manufactured by companies A, B, and C, have lifetime distributions EXP(1), EXP(2), and EXP(3), respectively. If bulbs are mixed in equal proportion in a box, find the probability that a randomly selected bulb will work for at least 1 year.
- 5.3.11** An oscillator sends the wave $X(t) = A \cos(2\pi t)$, where $A = 1$ or 2 with equal probabilities. We observe the value of $X(t)$ at the point chosen at random from the $U[n, n + 1]$ distribution for some n . Find: (i) $P(X(t) \leq 1)$. (ii) $P(|X(t)| > 3/2)$. (iii) $P(X(t) > 0)$.

5.4 FUNCTIONS OF RANDOM VARIABLES

In practical situations, we often deal with functions (or *transformations*) of random variables. Given the original distributions, we face the problem of determining the distributions of transformed random variables. The examples abound, starting from the simplest cases, such as a change of unit of measurement or a representation on the logarithmic scale. More complicated cases, discussed in Chapter 6, involve pairs of random variables such as conversion of a pair of Cartesian coordinates of a random point on the plane to polar coordinates of this point, or ratios of coordinates. The latter case occurs, for instance, in determining the distribution of velocity, calculated as the ratio of distance to time, both subject to variability (either in the actual measured velocity or due to errors of measurements).

Another compelling reason to study transformations of random variables could be the generation of random numbers that follow a specific distribution. A typical method here involves transformations. Any statistical software (or often even pocket calculator) can generate random numbers from distribution uniform on $[0, 1]$. Then, the desired random variables are obtained as suitable transformations of uniform random variables. Obviously to apply such a method, it is necessary to develop techniques of determining the distributions of functions of random variables, at least of those uniformly distributed.

Finally, the third reason is related to statistics. Observations such as experimental results and values recorded in the sample are regarded as values of random variables. These are often summarized into global indices, such as the average. Each such index (generally referred to as a *statistic*) is a function of the sample values, hence a function of random variables, and for statistical inference it is vital to know the distributions of such indices.

We begin with the conceptually and technically simplest case of transformations of one discrete random variable.

Let X assume values in the set $U = \{x_1, x_2, \dots\}$, with corresponding probabilities

$$p_i = P\{X = x_i\},$$

such that $\sum p_i = 1$. Then, $Y = \varphi(X)$, where φ is a real-valued function, also has a discrete distribution. The following example illustrates how the distribution of $Y = \varphi(X)$ can be obtained:

■ EXAMPLE 5.21

Suppose that variable X has the following distribution

Value	-2	-1	0	1	2	3	4
Probability	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$

If $\varphi(x) = x^2$, then $Y = X^2$, and the distribution of Y is

Value	0	1	4	9	16
Probability	$\frac{1}{10}$	$\frac{1}{10} + \frac{2}{10}$	$\frac{1}{10} + \frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$

Function φ is not one to one, and therefore

$$P\{Y = 4\} = P\{X^2 = 4\} = P\{X = 2\} + P\{X = -2\} = \frac{1}{10} + \frac{1}{10}.$$

$P\{Y = 1\}$ can be obtained similarly.

In general, we have

$$P\{Y = y\} = P\{\varphi(X) = y\} = P\{x : \varphi(x) = y\} = \sum_{x_i : \varphi(x_i) = y} P\{X = x_i\}.$$

An analogous formula can be used for functions of two variables, that is, random variables of the form $Z = \varphi(X, Y)$, where the distribution of Z is expressed in terms of the distribution of X and Y . The principle here is very simple, as can be seen in the next example.

■ EXAMPLE 5.22

Let the experiment consist of two tosses of a regular die, and let X and Y be the result on the first and second toss, respectively. We want to find the distribution of

$Z = \max(X, Y)$. Thus, the value of the function $\varphi(x, y)$ equals x or y depending on whether $x \geq y$ or $y \geq x$, and the random variable Z may be described as “the best out of two tosses.”

Clearly, Z may be 1, 2, $\dots$, 6, and it is simplest to find the distribution of Z by listing all of the outcomes associated with each specific value of Z . Thus, $Z = 1$ only if $(X, Y) = (1, 1)$; hence $P\{Z = 1\} = 1/36$. Next $Z = 2$ if the outcome is $(2, 1)$, $(1, 2)$, or $(2, 2)$ so that $P\{Z = 2\} = 3/36$. Proceeding in this way, we get

Z	1	2	3	4	5	6
Probability	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

It can be seen how “giving the second chance” improves the score. Larger values are much more likely than smaller values.

■ EXAMPLE 5.23

Let us continue Example 5.22, generalizing it to the case of n random variables (to be considered formally in Chapter 6). How large should n be to give a 99% or higher chance for the best of n tosses to equal at least 4?

SOLUTION. Let $X_1, X_2, \dots, X_n$ denote the result of consecutive tosses, and let $Z = \max(X_1, \dots, X_n)$. We want the smallest n with $P\{Z \geq 4\} \geq 0.99$. Let us therefore determine the distribution of Z , exhibiting its dependence of n . Direct enumeration was feasible for $n = 2$, as in the preceding example, but it is cumbersome for a larger n . It is easy to find $P\{Z \leq k\}$ for $k = 1, \dots, 6$. First, $\{Z \leq 1\} = P\{Z = 1\} = 1/6^n$, since only one outcome among the total of 6^n outcomes gives the maximum score of 1. Next

$$\{Z \leq 2\} = \{\max(X_1, \dots, X_n) \leq 2\} = \{X_1 \leq 2, X_2 \leq 2, \dots, X_n \leq 2\},$$

and we have

$$P\{Z \leq 2\} = \frac{2^n}{6^n}.$$

In general,

$$P\{Z \leq k\} = \frac{k^n}{6^n} \quad \text{for } k = 1, \dots, 6;$$

hence

$$P\{Z = k\} = P\{Z \leq k\} - P\{Z \leq k-1\} = \frac{k^n}{6^n} - \frac{(k-1)^n}{6^n}.$$

Since $P\{Z \geq 4\} \geq 0.99$ means that

$$P\{Z \leq 3\} = \left(\frac{3}{6}\right)^n \leq 0.01,$$

we have to solve the inequality $(1/2)^n \leq 0.01$. Taking $n \log(1/2) \leq \log 0.01$, we obtain

$$n \geq \frac{\log 0.01}{\ln(1/2)} = \frac{\log 100}{\log 2} = 6.64.$$

Thus, $n = 7$ tosses “practically guarantees” that at least one toss will lead to 4, 5, or 6.

We now consider the case of transformation of a single continuous random variable. Let F and f denote the cdf and density of X and let $Y = \varphi(X)$, where φ is assumed to be at least piecewise differentiable.

Regardless of whether we want the cdf or the density, the best strategy is to start by finding the cdf of Y . This method, occasionally referred to as the *cdf technique*, will be illustrated by some examples.

We begin with the simplest case, when φ is a strictly monotone function. If φ is increasing, we write for the cdf of Y :

$$F_Y(y) = P\{Y \leq y\} = P\{\varphi(X) \leq y\} = P\{X \leq \psi(y)\} = F_X(\psi(y)),$$

where ψ is the function inverse to φ .

The density is obtained by differentiating cdf, and consequently

$$f_Y(y) = \frac{d}{dy} F_X(\psi(y)) = f_X(\psi(y))\psi'(y). \quad (5.32)$$

If φ is monotonically decreasing, ψ must be a decreasing function too, and

$$F_Y(y) = P\{\varphi(X) \leq y\} = P\{X \geq \psi(y)\} = 1 - F_X(\psi(y)).$$

Therefore,

$$f_Y(y) = -f_X(\psi(y))\psi'(y), \quad (5.33)$$

a quantity that is positive, since now $\psi'(y) < 0$. Together, (5.32) and (5.33) lead to the following theorem:

Theorem 5.4.1 *If φ is a continuous differentiable function with inverse ψ and X is a continuous random variable with density f_X , then the density of $Y = \varphi(X)$ is*

$$f_Y(y) = f_X(\psi(y))|\psi'(y)|. \quad (5.34)$$

■ EXAMPLE 5.24 Linear Transformations

Let $Y = aX + b$. If $a > 0$, we can write

$$F_Y(y) = P\{aX + b \leq y\} = P\left\{X \leq \frac{y-b}{a}\right\} = F_X\left(\frac{y-b}{a}\right)$$

and

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right).$$

If $a < 0$, we have

$$F_Y(y) = P\{aX + b \leq y\} = P\left\{X \geq \frac{y-b}{a}\right\} = 1 - F_X\left(\frac{y-b}{a}\right)$$

and

$$f_Y(y) = -\frac{1}{a} f_X\left(\frac{y-b}{a}\right).$$

So for any $a \neq 0$,

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

The following transformation is important, both theoretically and practically. Let X have cdf F and density f , and let F be strictly increasing, so that F^{-1} exists. We consider the transformation $Y = F(X)$. Obviously, $0 < Y < 1$, so the distribution of Y is concentrated on $[0, 1]$. We therefore have $F_Y(y) = 0$ for $y \leq 0$ and $F_Y(y) = 1$ for $y \geq 1$, while for $0 < y < 1$ we write

$$F_Y(y) = P\{F(X) \leq y\} = P\{X \leq F^{-1}(y)\} = F(F^{-1}(y)) = y.$$

Consequently, $f_Y(y) = \frac{d}{dy}F_Y(y) = 1$ on $[0, 1]$, and we proved the next theorem.

Theorem 5.4.2 (Probability Integral Transform) *If X has continuous strictly increasing cdf F , then the distribution of $Y = F(X)$ is uniform on $[0, 1]$.*

This theorem may be formulated as follows: *If Y has $U[0, 1]$ distribution, then $X = F^{-1}(Y)$ has a distribution with the cdf F .* Thus, we have obtained a method of generating random variables with given continuous invertible cdf F , using random variables uniform on $[0, 1]$.

■ EXAMPLE 5.25

The assumption of monotonicity of φ (and hence also ψ) is crucial for the validity of (5.34), except for the obvious remark that what is really needed is monotonicity “in the domain where it really matters.” Specifically, if C is the set of all points x at which $f(x) > 0$ (called the *support* of X), then it suffices that φ is monotone only on C .

Consider the distribution with density

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0, \end{cases}$$

and let $\varphi(x) = x^2$. Then, φ is not monotone for all x , but it is monotone on the support of X , $C = [0, \infty)$. Thus, $\psi(x) = \sqrt{x}$, $\psi'(x) = 1/(2\sqrt{x})$, and the density of $Y = X^2$ is

$$f_Y(y) = \begin{cases} \frac{\alpha}{2\sqrt{y}} e^{-\alpha\sqrt{y}} & \text{for } y > 0 \\ 0 & \text{for } y \leq 0, \end{cases}$$

In the case where φ is not monotone, we still have

$$F_Y(y) = P\{Y \leq y\} = P\{\varphi(X) \leq y\},$$

but now φ has no inverse, and the inequality $\varphi(X) \leq y$ is usually not equivalent to a single inequality for X . Still the right-hand side can, in most cases, be represented through the cdf F_X evaluated at some points dependent on y . Differentiating, we can recover the density f_Y of Y .

This principle will now be illustrated by few examples.

■ EXAMPLE 5.26

Let X be a random variable with the density

$$f_X(x) = \begin{cases} 2.5 x^4 & \text{for } |x| \leq 1 \\ 0 & \text{otherwise.} \end{cases} \quad (5.35)$$

Let $\varphi(x) = x^2$, so that now φ is not monotone, and consider $Y = \varphi(X)$. Since $Y \geq 0$, we have $F_Y(y) = 0$ for $y \leq 0$, while for $y > 0$ we write

$$\begin{aligned} F_Y(y) &= P\{X^2 \leq y\} = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}). \end{aligned}$$

Differentiating, we get for $y > 0$,

$$f_Y(y) = F'_Y(y) = [f_X(\sqrt{y}) + f_X(-\sqrt{y})] \times \frac{1}{2\sqrt{y}},$$

so that (remembering that $f_X = 0$ if $x < -1$ or $x > 1$)

$$f_Y(y) = \begin{cases} 2.5 y^{3/2} & \text{for } 0 < y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

■ EXAMPLE 5.27 Square of a Normal Random Variable

Let X have a normal distribution with the density

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

and let $\varphi(x) = x^2$ so that now φ is not monotone. We have $F_Y(y) = 0$ for $y \leq 0$, while for $y > 0$ we write

$$F_Y(y) = P\{X^2 \leq y\} = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

Differentiating, we obtain

$$f_Y(y) = F'_Y(y) = [f_X(\sqrt{y}) + f_X(-\sqrt{y})] \times \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2} \quad (5.36)$$

for $y > 0$, and $f_Y(y) = 0$ for $y \leq 0$.

This is a special case of a *gamma* density, given by the formula $Cy^{a-1}e^{-by}$ ($y > 0$), where $a > 0$, $b > 0$, and C is the normalizing constant. Density (5.36) has $a = 1/2$, $b = 1/2$. Gamma distributions with $b = 1/2$ and a such that $2a$ is a positive integer form a family of *chi-square* distributions—very important in statistical inference. Both gamma and chi-square distributions will be discussed in more detail in following chapters.

■ EXAMPLE 5.28 Folded Normal Distribution

Let X be as in Example 5.27, and let now $\varphi(x) = |x|$. Then for $y > 0$,

$$F_Y(y) = P\{|X| \leq y\} = P\{-y \leq X \leq y\} = F_X(y) - F_X(-y),$$

and

$$f_Y(y) = f_X(y) + f_X(-y) = \sqrt{\frac{2}{\pi}} e^{-y^2/2}, \quad (5.37)$$

and $f_Y(y) = 0$ for $y \leq 0$. This distribution is sometimes also called *half-normal*.

PROBLEMS

- 5.4.1** If X is the result of tossing a balanced die, find the distribution of: **(i)** $Y = (X - 2)^2$. **(ii)** $Z = |X - 2.5|$.
- 5.4.2** Let X have the Poisson distribution with parameter λ , and let $Y = 2X$. Find the distribution of Y .
- 5.4.3** Let X have a continuous distribution with cdf F_X and density f_X , such that $F_X(0) = 0$. Find the density of variables: **(i)** $\sqrt{X}$. **(ii)** $\log X$. **(iii)** $1/X$. **(iv)** e^X .
- 5.4.4** Let X be $U[0, 1]$. Find φ such that $Y = \varphi(X)$ has $\text{EXP}(\lambda)$ distribution.
- 5.4.5** Let X have $\text{EXP}(\lambda)$ distribution, and let $Y = \sqrt{X}$. Find: **(i)** The cdf and density of Y . **(ii)** The lower quartile of Y .
- 5.4.6** Assume that X has the standard normal distribution. Find the density of: **(i)** $Y = X^3$. **(ii)** $Y = (X - 1)^3$. **(iii)** $Y = e^X$.
- 5.4.7** Find the density of $Y = X(1 + X)$ if X has $U[0, 1]$ distribution.
- 5.4.8** Random variable X has density $f_X(x) = Cx^4$ for $-2 \leq x \leq 1$ and 0 otherwise. Find the density of $Y = X^2$ (follow Example 5.26).
- 5.4.9** Let X have $U[-1, 1]$ distribution. Find the distribution of: **(i)** $Y = |X|$. **(ii)** $Z = 2|X| - 1$. **(iii)** Solve (i) and (ii) if the distribution of variable X is $U[-1, 2]$.
- 5.4.10** The duration (in days) of the hospital stay following a certain treatment is a random variable $Y = 4 + X$, where X has a density $f(x) = 32/(x + 4)^3$ for $x > 0$. Find: **(i)** The density of Y . **(ii)** The probability that a randomly selected patient will stay in the hospital for more than 10 days following the treatment.
- 5.4.11** Let X have density $f(x) = 2(1 - x)$, $0 \leq x \leq 1$. Find the density of: **(i)** $Y = X(1 - X)$. **(ii)** $W = \max(X, 1 - X)$.
- 5.4.12** Suppose that the measured radius R is a random variable with density

$$f_R(x) = \begin{cases} 12x^2(1 - x) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

In a circle with radius R , find the cdf of: **(i)** The diameter. **(ii)** The area.

- 5.4.13** The speed of a molecule of gas at equilibrium is a random variable X with density

$$f(x) = \begin{cases} kx^2 e^{-bx^2} & \text{for } x > 0 \\ 0 & \text{otherwise,} \end{cases}$$

where k is a normalizing constant and b depends on the temperature of the gas and the mass of the molecule. Find the probability density of the kinetic energy $E = mX^2/2$ of the molecule.

5.5 SURVIVAL AND HAZARD FUNCTIONS

In this section, we consider a special class of random variables that are particularly important in medical research (survival analysis) but also in engineering (reliability analysis) and social sciences.

These variables can serve as possible models for *lifetimes* of living organisms or of some equipment. They may denote the actual lifetime (i.e., time until death or equipment failure) or time until some event (e.g., recovery from a disease). Such random variables, denoted usually by T , are continuous and nonnegative.

Let F and f denote the cdf and density of T so that for $t \geq 0$,

$$F_T(t) = P\{T \leq t\} = \int_0^t f(u) du \quad (5.38)$$

[the integration starts at 0, since nonnegativity of T implies that $P\{T \leq 0\} = 0$, and hence $f(t) = 0$ for $t < 0$]. Moreover, let

$$S(t) = 1 - F(t) = P\{T > t\} = \int_t^\infty f(u) du \quad (5.39)$$

be the *survival distribution function* (or *survival function*) of the random variable T .

We introduce the following definition:

Definition 5.5.1 The function

$$h(t) = -\frac{d \ln S(t)}{dt}, \quad (5.40)$$

defined for $t > 0$ and $S(t) > 0$, is called the *hazard rate* (or *intensity rate*) *function* of random variable T . $\square$

By differentiating $-\ln(1 - F(t))$, it follows from (5.40) that at almost all points t we have

$$h(t) = \frac{f(t)}{S(t)}. \quad (5.41)$$

The following theorem expresses the cdf through the hazard function:

Theorem 5.5.1 If $h(t)$ is a hazard function of a random variable T , then its cdf equals

$$F(t) = 1 - e^{-\int_0^t h(u) du}.$$

Proof: The proof is immediate, by integrating (5.41) between 0 and t . $\square$

The interpretation of hazard functions is as follows: $h(t)\delta t$ can be approximated by

$$\frac{f(t)\delta t}{S(t)} \approx \frac{P\{t < T \leq t + \delta t\}}{P\{T > t\}};$$

hence, by the definition of conditional probability,

$$\begin{aligned} h(t)\delta t &\approx P\{t < T \leq t + \delta t | T > t\} \\ &= P\{\text{lifetime ends before } t + \delta t \mid \text{lifetime is longer than } t\}. \end{aligned}$$

In other words, $h(t)$ is the death rate at t of those who survived until t (are “at risk” at t). Thus, the hazard function describes the process of aging in terms of changes of risk of death with current age.

■ EXAMPLE 5.29

Let the random variable X have EXP(α) distribution. Then, $F(t) = 1 - e^{-\alpha t}$, $S(t) = e^{-\alpha t}$, and

$$h(t) = \frac{\alpha e^{-\alpha t}}{e^{-\alpha t}} = \alpha.$$

Thus, the exponential distribution describes the lifetime distribution in the case of lack of aging, when the risk of death does not change with age.

■ EXAMPLE 5.30

Suppose that you buy a new car, and let T be the time of the first breakdown. Typically, $h(t)$ is initially high, and then declines to a constant. It remains at this level for several years, eventually beginning to increase.

The reason is that early failures are typically caused by hidden faults of material, undetected in factory control. If they do not show up immediately, then there are probably no faults, and risk of failure remains constant for some time. The later increase is due to the wearing out of various parts.

In an engineering context, especially in problems of reliability of equipment, the hazard function is often called *failure rate* function.

Assuming that $F(t) < 1$ for all t (which is the most important case of interest), we can write

$$F(t) = 1 - e^{-H(t)}, \quad (5.42)$$

where $H(t) = -\log[1 - F(t)]$. On the other hand, in view of Theorem 5.5.1, we also have $H(t) = \int_0^t h(u) du$, which gives (5.24). Since $F(t) \rightarrow 1$ as $t \rightarrow \infty$, we must have

$$\int_0^\infty h(u) du = \infty. \quad (5.43)$$

Example 5.29 shows that the exponential distribution has a constant hazard, interpreted as lack of aging. If hazard is increasing, we have the phenomenon of aging (“new better than old”), while the opposite is true in case of decreasing hazard (“old better than new”). A flexible model of both situations is given by the following definition:

Definition 5.5.2 A nonnegative random variable T with the hazard rate

$$h(t) = Kt^\gamma, \quad t > 0, K > 0, \gamma > -1 \quad (5.44)$$

is said to have a *Weibull distribution*. □

Observe first that the condition $\gamma > -1$ ensures that the relation (5.43) holds so that T is a genuine random variable in the sense that

$$\lim_{t \rightarrow \infty} P\{T \leq t\} = 1$$

(i.e., $T = \infty$ is excluded). For reasons of tradition and convenience, one usually puts $\alpha = K/(\gamma + 1)$, $\beta = \gamma + 1$, so that the hazard rate of Weibull distribution takes the form

$$h(t) = \alpha\beta t^{\beta-1}, \quad \alpha > 0, \beta > 0, t > 0. \quad (5.45)$$

PROBLEMS

5.5.1 Find the hazard function of the $U[0, 1]$ distribution. Explain why $h(t)$ is unbounded.

5.5.2 Find the density and survival function of the distribution with hazard rate $h(t) = a + bt$ for $t > 0$, $a > 0$, $b > 0$.

5.5.3 Let X be a random variable with density

$$f(x) = \begin{cases} 0 & x < 0 \\ 0.5 & 0 \leq x \leq 1 \\ qe^{-\alpha x} & x > 1. \end{cases}$$

(i) Find q if α is known. (ii) Find hazard $h(t)$ and survival function $S(t)$.

5.5.4 Find the cdf and density of Weibull distribution with the hazard function (5.45).

5.5.5 Assume that the fuel pumps in a certain make of cars have lifetimes with a Weibull hazard rate $2/\sqrt[3]{t}$ (t measured in years). Find the probability that a fuel pump is still working after 5 months.

5.5.6 Let two distributions with densities f_1 and f_2 have constant hazard rates λ_1 and $\lambda_2 = a\lambda_1$, respectively. Show that $S_1(t) = S_2(t)^a$, if $S_1(t)$ and $S_2(t)$ are respective survival functions.

5.5.7 The series system is built in such a way that it operates only when all its three components operate (so it fails when at least one component fails). Assuming that the lifetime of each component has EXP (1) distribution and that the components operate independently, find the distribution and hazard function of the system's lifetime T .

5.5.8 A cancer specialist claims that the hazard function of random variable $T_c =$ "age at death due to cancer" is a bounded function which for large t has the form $h_c(t) = k/t^2$ (where t is the age in years and k is some constant).

Assume that $h(t)$ is the hazard of the time of death due to other reasons than cancer (other diseases, accidents, old age, etc.). If T_c and T are times of death with hazards $h_c(t)$ and $h(t)$, assume that the observed time of death is $T^* = \min(T_c, T)$, with $\max(T_c, T)$ being unobservable. To get an insight into the feasibility of the assumption of the cancer specialist in question, imagine that *all* reasons of death other than cancer have been eliminated (i.e., $H(t) \equiv 0$) and find the probability that a person will live forever.